BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Pseudocode and Trace Tables

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What does a diamond represent in a flowchart?

2. Why is primary storage needed?
   A) To hold data in use
   B) To display images
   C) To connect network devices
   D) To print documents

3. "Both RAM and ROM lose their contents when power is removed." Correct this. Explain why.

4. A computer needs startup instructions when switched on. "ROM normally holds all unsaved user work while applications run." Correct this. What could happen?

Today's objectives

Read and explain pseudocode statements. Trace changing variable values in the correct order.

Key vocabulary - Read, cover, recall

Pseudocode
Readable structured instructions describing an algorithm.

Trace table
Table following variable values and outputs during execution.

Variable
Named storage whose value can change.

Assignment
Storing a value in a variable.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Pseudocode represents an algorithm

Pseudocode expresses steps in clear, structured language. OCR Exam Reference Language gives a consistent notation for variables, selection and output.

Record each assignment

A trace table follows execution instruction by instruction. After assignment, record the new value of the variable; do not use its old value again.

Trace conditions accurately

Evaluate an IF condition against the latest value. Only execute the true or false branch selected. Record printed output when a print statement runs.

Find the first incorrect step

Compare the table with expected values. When the first mismatch appears, correct that instruction and repeat the dry run with the same inputs.

I Do - Worked model

Trace: total = 4; total = total + 3; print(total). State total after each assignment and the final output. [3 marks]

- The first line sets total to 4. Use the current value for later statements.
- After total = 4, record 4 [1].
- Next, total = total + 3 uses 4 + 3, so total becomes 7 [1].
- print(total) outputs 7 [1], but does not change the variable.

We Do - Guided practice

1. Trace: score = 5; score = score + 2; print(score). Give both values and the output. [3 marks]

2. Trace: count = 9; count = count - 4; if count > 6 then print("High") else print("Low") endif. State count and output. [4 marks]

3. A pupil expects 12 but a trace shows 10. Explain how a trace table helps find and fix the error. [4 marks]

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Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the number printed after two instructions.

```
storedMB = 10
storedMB = storedMB + 4
print(storedMB)
```

You Do - Independent exam questions

Q2. State one use of a trace table. [2 marks]

Q3. Trace: cost = 8; cost = cost * 2; cost = cost + 3; print(cost). State cost after each assignment and the output. [4 marks]

Q4. Trace: balance = 20; balance = balance - 6; if balance >= 15 then print("Enough") else print("Low") endif. Explain the branch and output, including the changes to balance. [6 marks]

Exit Ticket - True or False?

The latest value of a variable is used when tracing the next instruction.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. A decision or condition with alternative paths.

2. A - To hold data in use

3. RAM loses its contents; ROM keeps its contents because it is non-volatile.

4. ROM keeps firmware or startup instructions without power. These let the computer start when switched on.

I Do - Explain the model

The first line sets total to 4. Use the current value for later statements.

After total = 4, record 4 [1].

Next, total = total + 3 uses 4 + 3, so total becomes 7 [1].

print(total) outputs 7 [1], but does not change the variable.

We Do - Model answers

1. First score is 5 [1]. Then score becomes 7 [1]. Output is 7 [1].

2. First count is 9 [1]; after subtraction it is 5 [1]. 5 > 6 is false [1]; Low is printed [1].

3. Records each variable's value after executed steps [1], allowing comparison with expected values [1]. The first incorrect value identifies where to inspect [1]; correct and rerun the trace [1].

Code First - Answer

```
storedMB = 10
storedMB = storedMB + 4
print(storedMB)
```

Expected output: 14

You Do - Self-mark

Q2: Records changing variable values as instructions execute [1], helping predict output/check logic [1].

Q3: Cost 8 [1], then 16 [1], then 19 [1]; output 19 [1].

Q4: Balance starts 20 [1], decreases to 14 [1]. 14 >= 15 is false [1] because 14 is below 15 [1]. The else branch executes [1], printing Low [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.